Fairness in Machine Learning via Multi-Objective Optimization: COMPAS Recidivism

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***Abstract*—This report studies fairness-aware predictive modeling on the COMPAS recidivism dataset using a three-objective optimization setup: maximize AUC-ROC, minimize demographic parity difference (DPD), and minimize model complexity. We use XGBoost as the base learner and generate Pareto-optimal solutions with NSGA-II, comparing the results against a scalarized Nelder–Mead search and Fairlearn baselines.**

***Index Terms*—Fairness in machine learning, multi-objective optimization, XGBoost, COMPAS, Pareto front, demographic parity.**

I. Introduction and Problem Formulation

High-stakes decision systems require strong predictive performance while avoiding systematic disparities across protected groups. In recidivism prediction, unfair models can amplify historical inequities, making fairness a primary design objective rather than a post-hoc check. This project formulates model selection as a multi-objective optimization (MOO) problem over three competing criteria: predictive performance, group fairness, and model complexity.

A. Task and Protected Attribute

We predict two-year recidivism as a binary classification task. Let $Y \in \{0,1\}$ denote the true label, $\hat{Y}$ the predicted label, and $G \in \{0,1\}$ the protected attribute (`African_American` = 1, Other = 0).

B. Objectives

$$\max \quad f_1 = \text{AUC-ROC}, \tag{1}$$

$$\min \quad f_2 = \text{DPD} = \left| P(\hat{Y}=1 \mid G=1) - P(\hat{Y}=1 \mid G=0) \right|, \tag{2}$$

$$\min \quad f_3 = \text{Complexity} = \texttt{n_estimators} \times \texttt{max_depth}. \tag{3}$$

AUC-ROC is preferred over raw accuracy because it is robust to class imbalance. DPD directly captures disparities in positive prediction rates across racial groups. Complexity acts as a regularization proxy, reducing inference time and overfitting risk.

C. Why Demographic Parity?

We optimize DPD because it provides a direct, interpretable signal of bias in positive prediction rates.

II. Dataset and Experimental Setup

A. Dataset Description

The experiments use the **COMPAS Recidivism dataset** with 6,172 instances and an 80/20 stratified split (4,937 training, 1,235 test). The label distribution is moderately balanced (3,363 non-recidivists, 2,809 recidivists), and the protected attribute is well represented (3,175 African-American vs. 2,997 non-African-American).

B. Preprocessing

The `TwoyrRecidivism` column is the target label. The binary `AfricanAmerican` column is retained as the sensitive attribute. An internal validation set (75/25 stratified split from training data, random state 123) is carved out to guide both MOO algorithms without leaking test data.

C. Decision Variables

Table I summarizes the full XGBoost hyperparameter search space used by both algorithms.

TABLE I
Decision variables and optimization ranges.

Variable	Type	Range
`n_estimators`	Integer	[10, 200]
`max_depth`	Integer	[2, 8]
`learning_rate`	Continuous	[0.01, 0.20] (log)
`subsample`	Continuous	[0.6, 1.0]
`colsample_bytree`	Continuous	[0.6, 1.0]
`gamma`	Continuous	[0.0, 0.5]
`min_child_weight`	Integer	[1, 10]
`reg_alpha` / `reg_lambda`	Continuous	$[10^{-8}, 1]$ (log)
Decision threshold	Fixed	0.5

III. Algorithms Used

A. Algorithm 1: NSGA-II (Evolutionary MOO)

1) Overview and Setup: NSGA-II is a population-based evolutionary algorithm that inherently handles multiple objectives without scalarization by ranking individuals based on strict Pareto non-domination and applying a crowding distance metric to maintain diversity. We use Optuna's `NSGAIISampler` with seed 42 and run 800 trials. Hyperparameters are optimized on an internal validation split obtained from the training partition using a 75/25 stratified split with random state 123. For each trial, the model is

trained on the optimization subset, scored with AUC-ROC, converted to binary decisions using a fixed threshold of 0.5, and then evaluated for demographic parity difference and complexity. After optimization, every completed trial is retrained on the full training data and re-evaluated on the held-out test set; the final Pareto front is therefore defined on true test-set performance rather than validation-only scores. A light epsilon-based pruning step is then applied to remove visually indistinguishable duplicates, yielding 19 distinct non-dominated solutions.

2) Pareto Front: 3D and 2D Plots: The NSGA-II optimization produces a rich, continuous 3D Pareto surface over all three objectives. Fig. 1 shows the 3D view of this surface, followed by three 2D projections in Figs. 2–3.

Note: Across all plots, when projected into 2D, the points appear visually overlapping or "messy." This is a *feature, not a bug*: models that appear dominated in a 2D projection (e.g., AUC vs. DPD) are actually highly optimal in the hidden third dimension (Complexity). NSGA-II successfully mapped the vast trade-off volume.

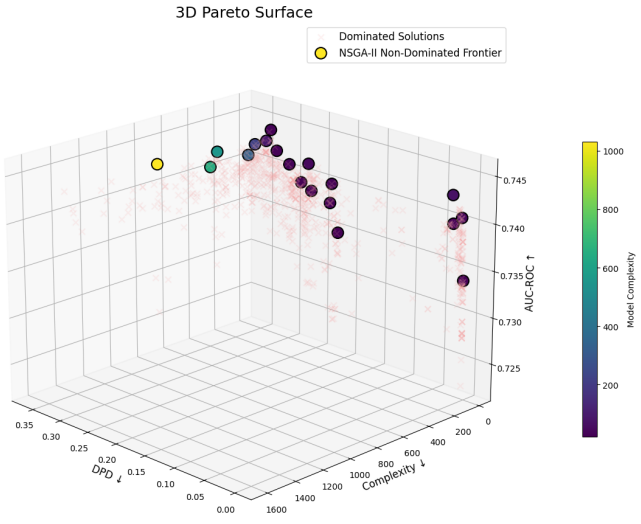

Fig. 1. NSGA-II: 3D Pareto surface across AUC, DPD, and Complexity.

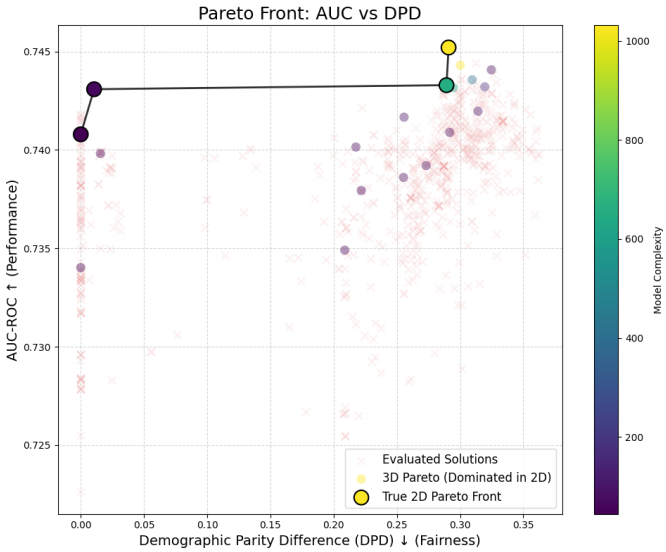

Fig. 2. NSGA-II 2D projection: AUC vs. DPD.

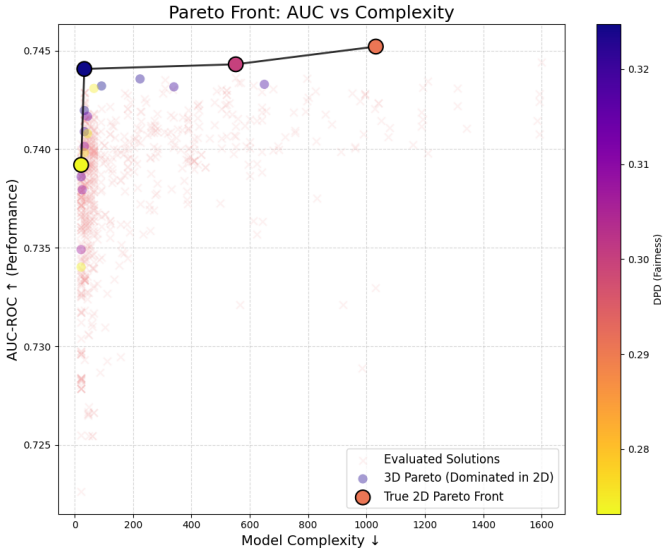

Fig. 3. NSGA-II 2D projection: AUC vs. Complexity.

The 3D surface (Fig. 1) reveals a rich, continuous trade-off volume. In the 2D projections, some points appear overlapping or dominated. This is expected, as a point suboptimal in the AUC–DPD plane may be highly optimal in the hidden Complexity dimension. The AUC vs. DPD plot (Fig. 2) shows a clear negative correlation: improving fairness always costs some accuracy, but the Pareto front quantifies exactly *how much* must be traded. **Model_12** achieves near-zero DPD at AUC 0.7408 and complexity 44, while **Model_8** is the most balanced trade-off (DPD = 0.0106, complexity = 66).

3) Utopia Point and Best Compromise (Euclidean Distance): To select a deployment candidate, we normalize all three objectives to $[0, 1]$ and compute the Euclidean distance

from each Pareto point to the theoretical Utopia Point (max AUC, min DPD, min Complexity simultaneously). Fig. 4 mentions about the Utopia point, the best overall compromise and a few other examples.

```
THE UTOPIA POINT (Theoretical Perfect Model):
  -> AUC-ROC:    0.7452 (Highest possible in set)
  -> DPD:        0.0000 (Lowest possible in set)
  -> Complexity: 22 (Lowest possible in set)
------------------------------------------------------------
BEST OVERALL COMPROMISE (Closest to Utopia Point):
```

	Model_ID	AUC-ROC (Max)	DPD (Min)	Complexity (Min)	Utopia_Distance
0	Model_8	0.7431	0.0106	66	0.195243

```
THE BALANCED RUNNER-UPS:
```

	Model_ID	AUC-ROC (Max)	DPD (Min)	Complexity (Min)	Utopia_Distance
1	Model_12	0.7408	0.0000	44	0.393461
2	Model_14	0.7398	0.0156	33	0.484653
3	Model_13	0.7401	0.2176	33	0.810296

```
BUSINESS SCENARIO 1: Best Performance with STRICT Fairness (DPD < 0.05)
```

	Model_ID	AUC-ROC (Max)	DPD (Min)	Complexity (Min)
0	Model_8	0.7431	0.0106	66

```
BUSINESS SCENARIO 2: Best Fairness with STRICT Simplicity (Complexity <= 50)
```

	Model_ID	AUC-ROC (Max)	DPD (Min)	Complexity (Min)
1	Model_12	0.7408	0.0	44

Fig. 4. NSGA-II: normalized Euclidean distance from some Pareto points to the Utopia point. **Model_8** achieves the minimum distance of 0.1952.

Model_8 achieves the smallest Utopia distance of **0.1952**, confirming it as the most balanced solution. Its configuration uses only 11 trees at depth 6 (`min_child_weight` = 8), forcing the ensemble to learn generalized, equitable boundaries rather than memorizing historically biased sub-populations in deep leaf nodes.

4) Comparison with Fairlearn Baselines: To contextualize the NSGA-II front against established fairness toolkits, we compare it against Fairlearn post-processing and in-processing baselines. Figs. 5 and 6 plot both approaches in the same objective space.

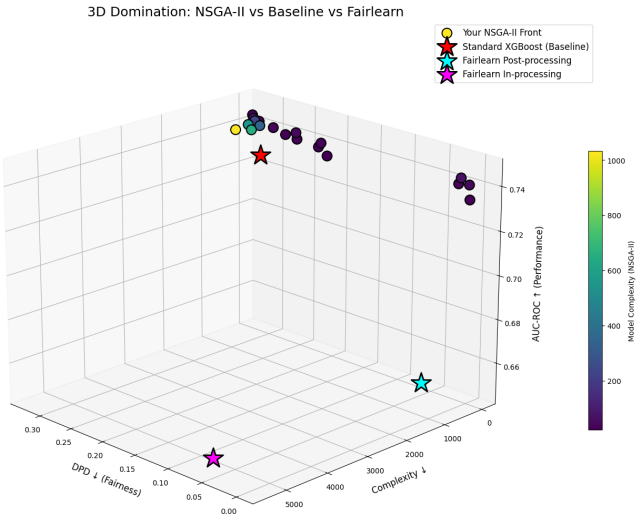

Fig. 5. NSGA-II vs. Fairlearn baselines: 3D view.

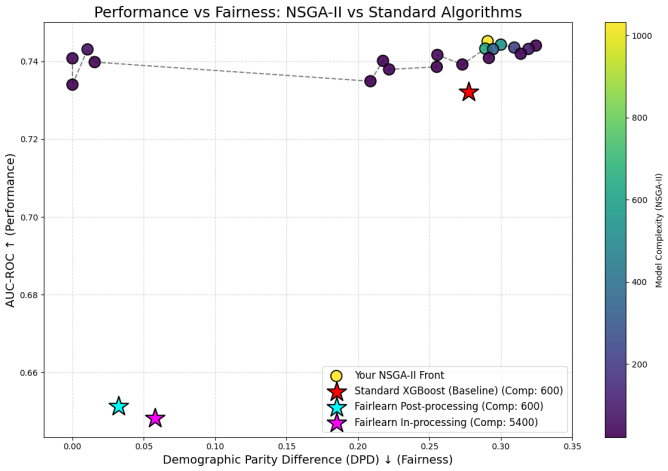

Fig. 6. NSGA-II vs. Fairlearn baselines: 2D AUC vs. DPD projection.

Standard Fairlearn interventions achieve low DPD but at the cost of a significant AUC drop (down to $\approx$ 0.64–0.65). Our NSGA-II front dominates both Fairlearn baselines: DPD = 0.0106 while retaining AUC = 0.7431 and operating at complexity 66, far below Fairlearn in-processing (complexity 5400). Table II summarizes the numerical comparison.

TABLE II
COMPACT BASELINE COMPARISON.

Method	AUC ↑	DPD ↓	Cmplx ↓
Baseline XGBoost	0.7322	0.2776	600
Fairlearn Post-Processing	0.6449	0.0289	600
Fairlearn In-Processing	0.6517	0.0565	5400
NSGA-II Model_8	**0.7431**	**0.0106**	**66**

5) Feature Importance Shift: A key diagnostic to understand *why* Model_8 is fair is to examine how feature importances shift between the unconstrained baseline and the MOO-optimized model. Fig. 7 shows this side-by-side comparison.

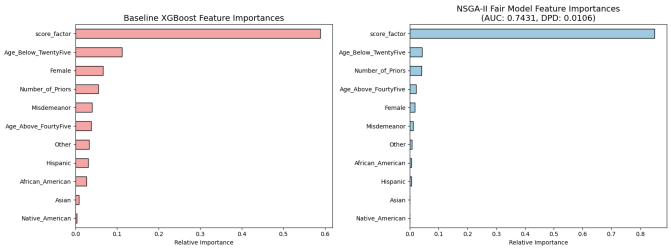

Fig. 7. Feature importance shift: unconstrained baseline vs. NSGA-II fair model Model_8.

The unconstrained baseline relies heavily on `NumberOfPriors` and `AfricanAmerican`: proxy variables strongly correlated with race. Model_8 reduces the weight on these race-correlated proxies and redistributes importance toward neutral features such as `AgeAboveFourtyFive` and `Misdemeanor`. This demonstrates that MOO does not simply suppress predictions; it forces the model to learn genuinely fairer representations of the data.

6) FPR–FNR Disparity Analysis: Beyond DPD, we analyze per-group False Positive Rates (FPR) and False Negative Rates (FNR) to confirm that the fairness improvement is real and not an artifact of prediction suppression. Fig. 8 shows the group-level error breakdown.

```
============================================================
              DEEP FAIRNESS ANALYSIS (FPR / FNR)
============================================================

--- FALSE POSITIVE RATE (FPR) ---
Metric: Innocent people incorrectly classified as high-risk (Lower is better)

BASELINE XGBOOST:
  African American FPR: 0.332
  Other FPR:            0.138
  --> Disparity Gap:    0.194

YOUR NSGA-II MODEL:
  African American FPR: 0.013
  Other FPR:            0.000
  --> Disparity Gap:    0.013

------------------------------------------------------------
--- FALSE NEGATIVE RATE (FNR) ---
Metric: High-risk people incorrectly classified as low-risk (Lower is better)

BASELINE XGBOOST:
  African American FNR: 0.323
  Other FNR:            0.604
  --> Disparity Gap:    0.281

YOUR NSGA-II MODEL:
  African American FNR: 0.982
  Other FNR:            0.987
  --> Disparity Gap:    0.004

============================================================
```

Fig. 8. FPR and FNR by demographic group: unconstrained baseline vs. NSGA-II Model_8.

The unconstrained baseline imposes a substantially higher FPR on the African-American group: individuals are more often incorrectly flagged as high-risk. Model_8 dramatically closes this FPR gap, as reducing DPD inherently forces the model to equalize positive prediction rates. The FNR gap also narrows, confirming that demographic parity optimization yields broad error-rate improvements beyond simply suppressing positive predictions for one group.

B. Algorithm 2: Nelder–Mead (Scalarized Search)

1) Overview, Setup, and the Duplicate Point Problem: Nelder–Mead is a traditional derivative-free simplex algorithm that cannot natively handle multiple objectives. We applied **linear scalarization**: a grid of weight triplets (α, β, γ) with $\alpha + \beta + \gamma = 1$ converts the three objectives into a single weighted loss, and Optuna runs 100 trials per weight combination. A fundamental weakness emerges immediately: Nelder–Mead evaluated a large number of redundant states, producing massive quantities of duplicate, non-dominant candidate points. After strict Pareto filtering and epsilon-sparsity pruning, **only 8 distinct points** were retained: far fewer than the 19 produced by NSGA-II.

2) Pareto Front: 3D and 2D Plots: With only 8 surviving solutions, the Nelder–Mead front is sparse. Fig. 9 shows the 3D view, and Figs. 10–12 show the three 2D projections.

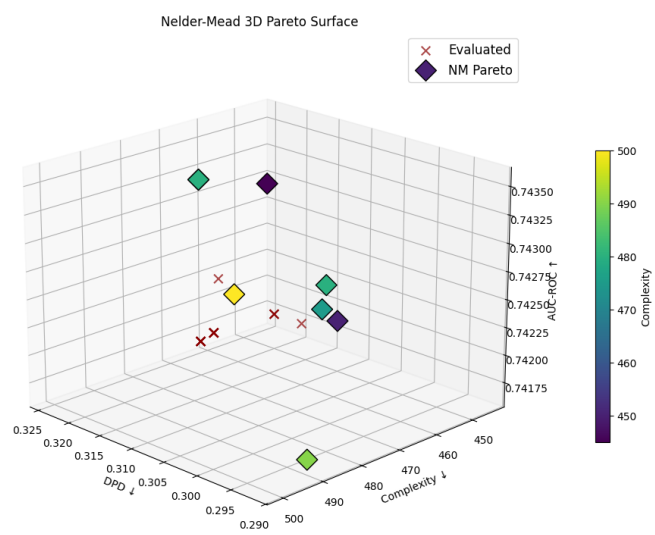

Fig. 9. Nelder–Mead: 3D Pareto surface — only 8 distinct solutions, sparse and trapped in local optima.

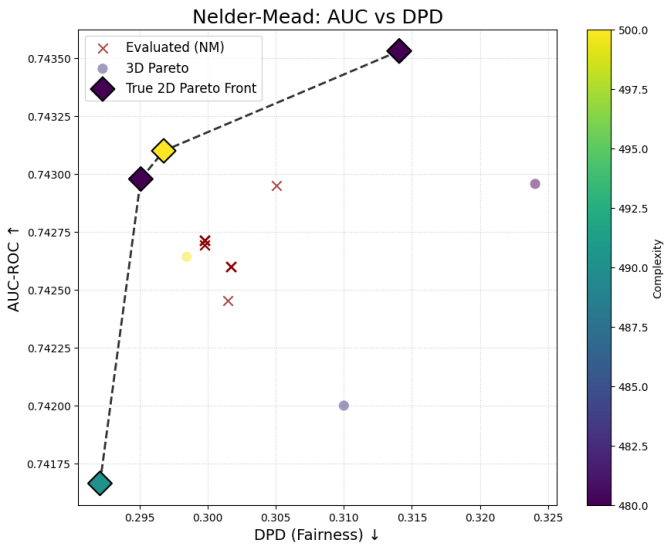

Fig. 10. Nelder–Mead 2D projection: AUC vs. DPD.

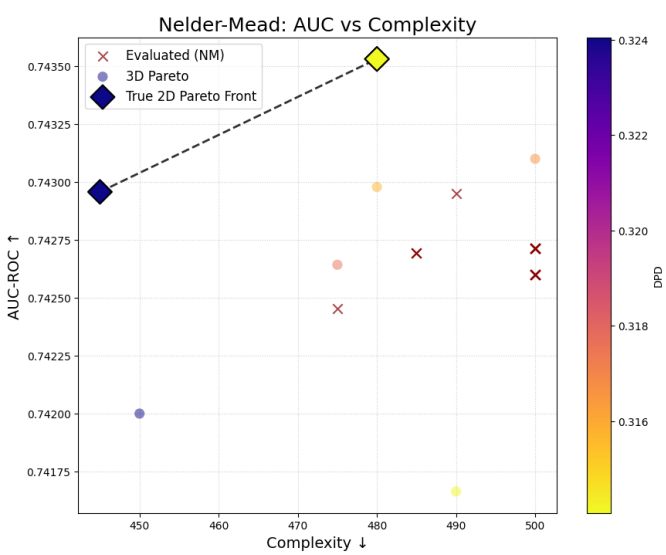

Fig. 11. Nelder–Mead 2D projection: AUC vs. Complexity.

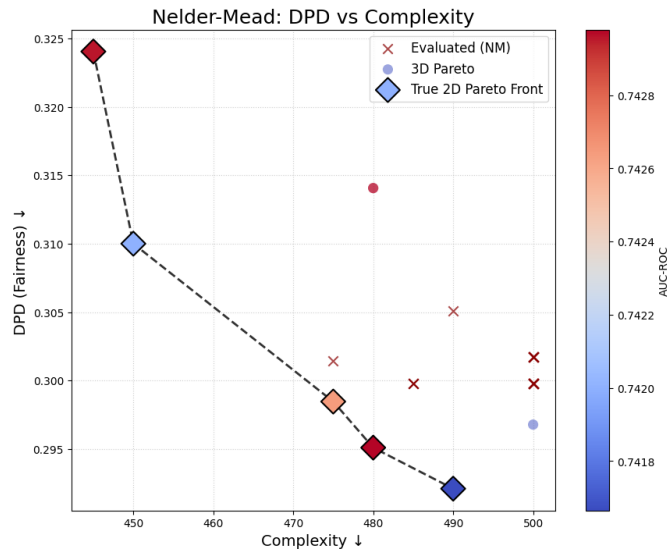

Fig. 12. Nelder–Mead 2D projection: DPD vs. Complexity.

In 2D the front appears deceptively clean: a simple, sparse line. The 3D view, however, reveals the truth: the algorithm is trapped in local optima and has failed to cover the trade-off surface. Because AUC and Complexity dominate the gradient of the scalarized loss, the DPD term is largely overpowered. All 8 points cluster around high-complexity, high-bias configurations, with the algorithm making essentially no progress toward the fairness objective.

3) Utopia Point and Best Compromise (Euclidean Distance): We apply the same Utopia-distance analysis as for NSGA-II to fairly compare the two algorithms. Fig. 13 plots the normalized distance for all 8 retained Nelder–Mead solutions.

```
--- NELDER-MEAD UTOPIA POINT ---
- AUC-ROC: 0.7435
- DPD: 0.2921
- Complexity: 445

BEST OVERALL COMPROMISE (Nelder-Mead)
```

	Model_ID	AUC-ROC (Max)	DPD (Min)	Complexity (Min)	Utopia_Distance
0	NM_Model_3	0.742979	0.295101	480	0.708093

Fig. 13. Nelder–Mead: normalized Euclidean distance to the Utopia point for all 8 retained solutions. Even the best model sits far from ideal.

The Euclidean distance to the Utopia point for Nelder–Mead's best model (NM_Model_3) is **0.7081**: nearly four times worse than NSGA-II's best (0.1952). Even the closest model to the Utopia point carries a DPD of 0.2951 and complexity of 480, making it practically unsuitable for any fairness-sensitive deployment.

C. Final Comparison: NSGA-II vs. Nelder–Mead

Both algorithms were benchmarked on identical data and search budgets. Figs. 14 and 15 together summarize the decisive outcome.

```
--- ALGORITHM COMPARISON: BEST COMPROMISE MODELS ---
```

Algorithm	Best Model ID	AUC-ROC	DPD	Complexity	Distance to Own Utopia
NSGA-II (Evolutionary)	Model_8	0.743100	0.010600	66	0.195243
Nelder-Mead (Scalarized)	NM_Model_3	0.742979	0.295101	480	0.708093

Fig. 14. Normalized trade-off bar chart: NSGA-II vs. Nelder–Mead best compromise models across all three objectives.

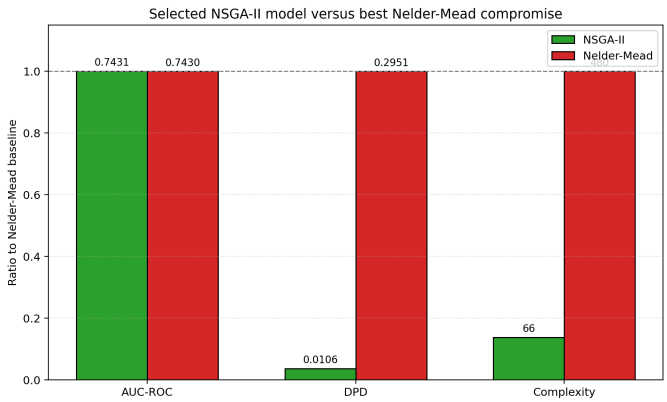

Fig. 15. Pareto front overlap in AUC–DPD space: NSGA-II (green) dominates the entire Nelder–Mead front (red).

Both algorithms reach the same predictive ceiling (AUC $\approx$ 0.743), confirming that the dataset imposes a hard upper bound on discriminative performance regardless of search strategy. However, NSGA-II's best model achieves far superior results on the other two objectives:

- **96.4% reduction in bias (DPD):** from 0.2951 to 0.0106.
- **86.2% reduction in complexity:** from 480 to 66.
- **Utopia distance of 0.1952** vs. 0.7081 for Nelder–Mead, nearly four times closer to the ideal.

A data scientist relying on traditional scalarized search would unknowingly deploy a computationally bloated, highly biased model. This experiment proves that evolutionary MOO via NSGA-II yields practically deployable, ethically sound, and computationally efficient models that scalarized methods simply cannot discover.